MAP 001:

The Second Loop

Worlds for Continual Machine Development

Agents as World

A computational organization for persistent worlds through which physical agents can continue to develop.

Machine Aliveness Position 001 / September 2026

ABSTRACT

Machine Aliveness studies machines that act with initiative, remain grounded in a body, retain history, and live alongside people. This paper examines one part of that agenda: how such machines can continue to develop through experience in physical and social worlds that do not reset around them. A machine that lives alongside people does not encounter a sequence of independent tasks. It accumulates history, changes the conditions of later interaction, and returns to a world already altered by what happened before.

Reality is therefore indispensable, but insufficient as the only substrate of development. It provides committed consequences and external evidence, yet it cannot be freely forked, replayed, accelerated, or restored. A household cannot be duplicated from the same instant; a person cannot be reset to a prior experience; months of interaction cannot be rerun under controlled counterfactuals.

We call the missing computational layer the **Second Loop**. The First Loop unfolds in reality. The Second Loop makes the conditions of future development computational: lived experience can inform world reconstructions; candidate interactions can be branched and compared; models of people, processes, and environments can be revised; and the machinery that constructs later developmental situations can itself be retained and improved. The Second Loop does not replace reality. It expands what can be

learned from it while preserving the distinction between internally generated evidence and external experience.

We propose **Agents as World (AAW)** as an organization for this computational world. The underlying object is a **World Process**: a persistent computational process through which participants, events, physical consequences, and histories continue to unfold. Actor Agents participate from local perspectives; World Agents construct, coordinate, and revise ongoing processes; and a tool-grounded World Kernel acts as a commitment layer, separating proposed futures from committed history.

For human-centered physical intelligence, people are not merely sources of demonstrations, instructions, or feedback. They are persistent participants whose memory, intentions, routines, and relationships evolve through interaction and become part of the world a machine encounters again.

This is one route from machine intelligence toward machine aliveness: not simply more capability, but continued development across a lived history, supported by computational worlds that can remember, branch, change, and remain grounded in reality.

> THE POSITION
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> The First Loop unfolds within a world. The Second Loop revises the processes through which later worlds unfold.

1. Development beyond the episode

1.1 The world continues

Embodied intelligence is often organized around episodes: a task begins, an agent acts, a trajectory ends, and learning proceeds from what happened. But a machine that lives in the real world does not encounter the world this way.

The world continues. Objects remain moved. People remember. Routines shift. Expectations form. The machine itself carries forward new memories, models, tools, and behaviors. What happens now changes the conditions of what can happen next.

We use **development** for this broader process of persistent change. Learning is part of it, but development also includes changes in memory, interaction strategy, models of people and environments, and the machinery through which future experience is produced.

1.2 The First Loop is reality

For physical intelligence, reality is the First Loop: the place where actions acquire real consequences and internal assumptions meet external evidence.

But reality does not branch easily. A household cannot be forked from the same moment; a person cannot be restored to an earlier experience; a month of interaction cannot be replayed under alternative conditions. Physical time, safety, hardware, and human participation make systematic counterfactual exploration expensive.

Reality is therefore the most authoritative source of evidence, but a poor substrate for exploring every possible path of development.

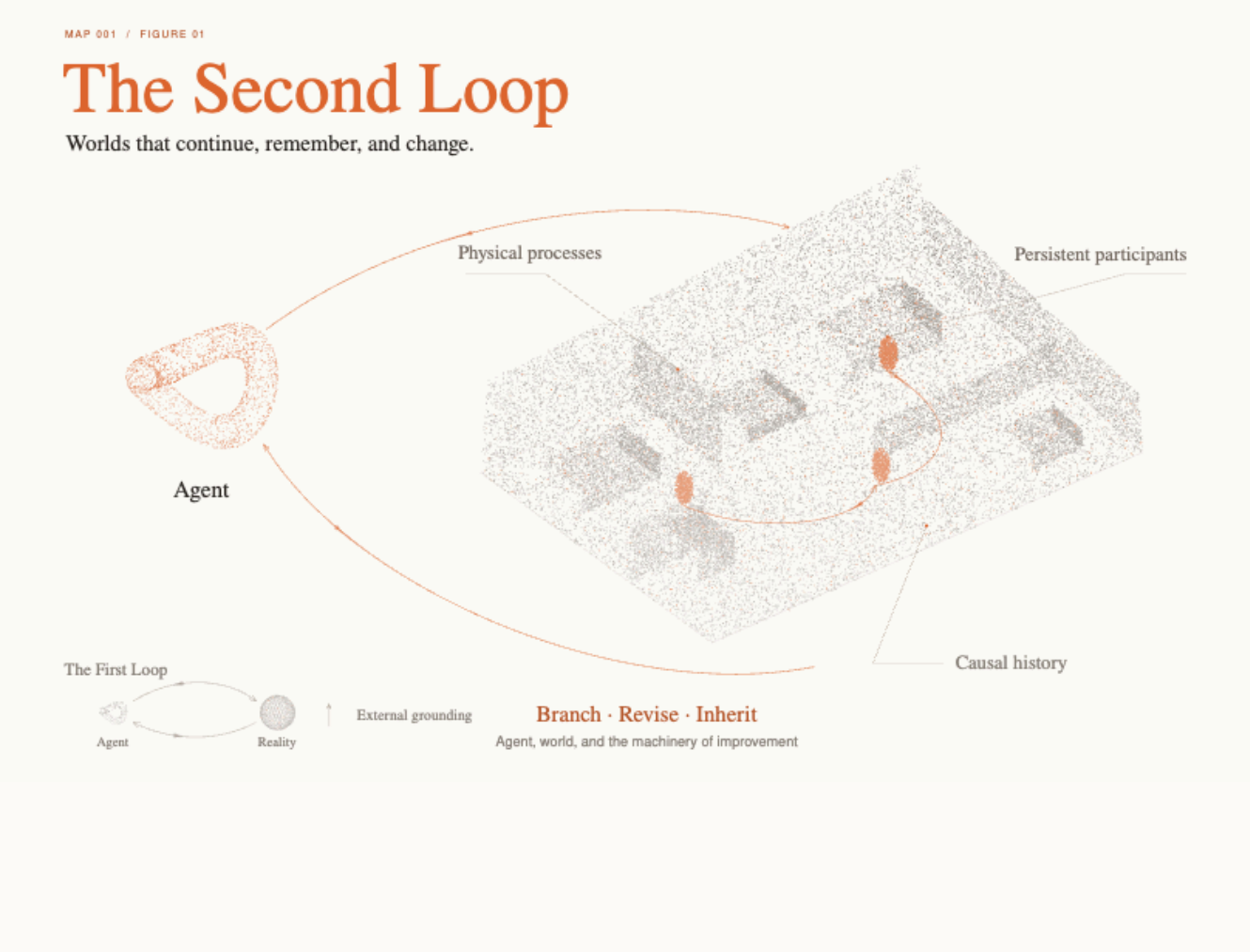
1.3 The Second Loop

We call the missing computational layer the **Second Loop**.

The First Loop is where consequences become real. The Second Loop is where the conditions of future development become computational.

Within it, lived experience can be reconstructed, alternative futures can be explored, models of people and environments can be revised, and the processes that generate later developmental situations can themselves change.

The Second Loop does not replace reality. It expands what can be learned from it.



2. The world as a continuing process

2.1 The world side must be executable and revisable

Once the world becomes part of a machine's developmental process, it is no longer adequately described as a ~~passive environment~~ or a ~~predictor of observations.~~ We call the object of interest a **World Process**: a persistent computational process through which participants, events, physical consequences, and histories continue to unfold over time.

A World Process specifies more than what exists. It determines what continues when nobody is observing, what each participant can perceive, how physical and human dynamics interact, how time advances, and which mechanisms can be revised between world versions. Its implementation may combine neural models, programs, scene descriptions, process models, actor specifications, physics settings, state stores, and executable tools.

It need not be end-to-end differentiable. What matters is that the world can **run**, that its consequences can be inspected, and that the machinery governing later worlds can be revised.

Revision, however, does not make ~~history negotiable.~~ The Second Loop may change the mechanisms through which later worlds are instantiated, but a world already underway must preserve its committed consequences. A revision creates a new world version; it does not ~~retroactively rewrite~~ what participants have already observed. This tension—**revisable processes, committed history**—will motivate the World Kernel in Section 3.

2.2 The human is inside the loop

Generalist embodied systems already use humans extensively—as sources of human video and demonstrations, and as instruction-givers or interactive operators at execution time. [13] [14] These are essential forms of human grounding. In these roles, however, the person largely appears at the boundary of the learning system: supplying data, commands, corrections, or evaluation.

For machines that develop through repeated interaction with people, that boundary is too narrow. People are part of the world dynamics the robot acts within, and part of the dynamics that remain after an interaction ends. A household member does not reset after a task. They remember interactions, form expectations, carry private information, change routines, become more or less receptive, and bring those changes into later encounters.

> THE HUMAN BOUNDARY
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> People are not merely interfaces to the world. They are persistent, adaptive parts of the world.

This changes the modeling role of the human. A demonstrator provides data; an instructor provides a command; an evaluator provides a judgment. A **Human Agent** inside the world instead carries state forward and participates in the causal history through which later interaction unfolds. Persistent social agents and shared human–robot worlds have precedents in Generative Agents and Virtual Community. [5] [6]

Our concern is not simply whether a model can predict what a person will do next, but how persistent human state becomes part of the developmental world a machine repeatedly encounters.



2.3 The world remembers and commits

A continuing world carries its consequences forward. Objects remain changed. People remember. Routines shift. Ongoing processes constrain what can happen next. The learner therefore does not simply act in a world; its actions partly shape the world it will encounter again.

This persistence is causal, not merely archival. A physical action may change what a person can observe; an interruption may change attention; a social response may alter the learner's next action; and that action may in turn change both the physical scene and a later interaction. Physical, human, and interaction dynamics unfold as one history.

For those consequences to matter, the world must preserve enough of that history to make them available to the future. At the same time, it must distinguish **possible futures from realized ones**. Proposed events may branch or be revised; consequences that have already occurred become part of committed history.

The Second Loop may change how later worlds unfold. It should not silently change the world from which they unfold.

2.4 World-making must expose its own machinery

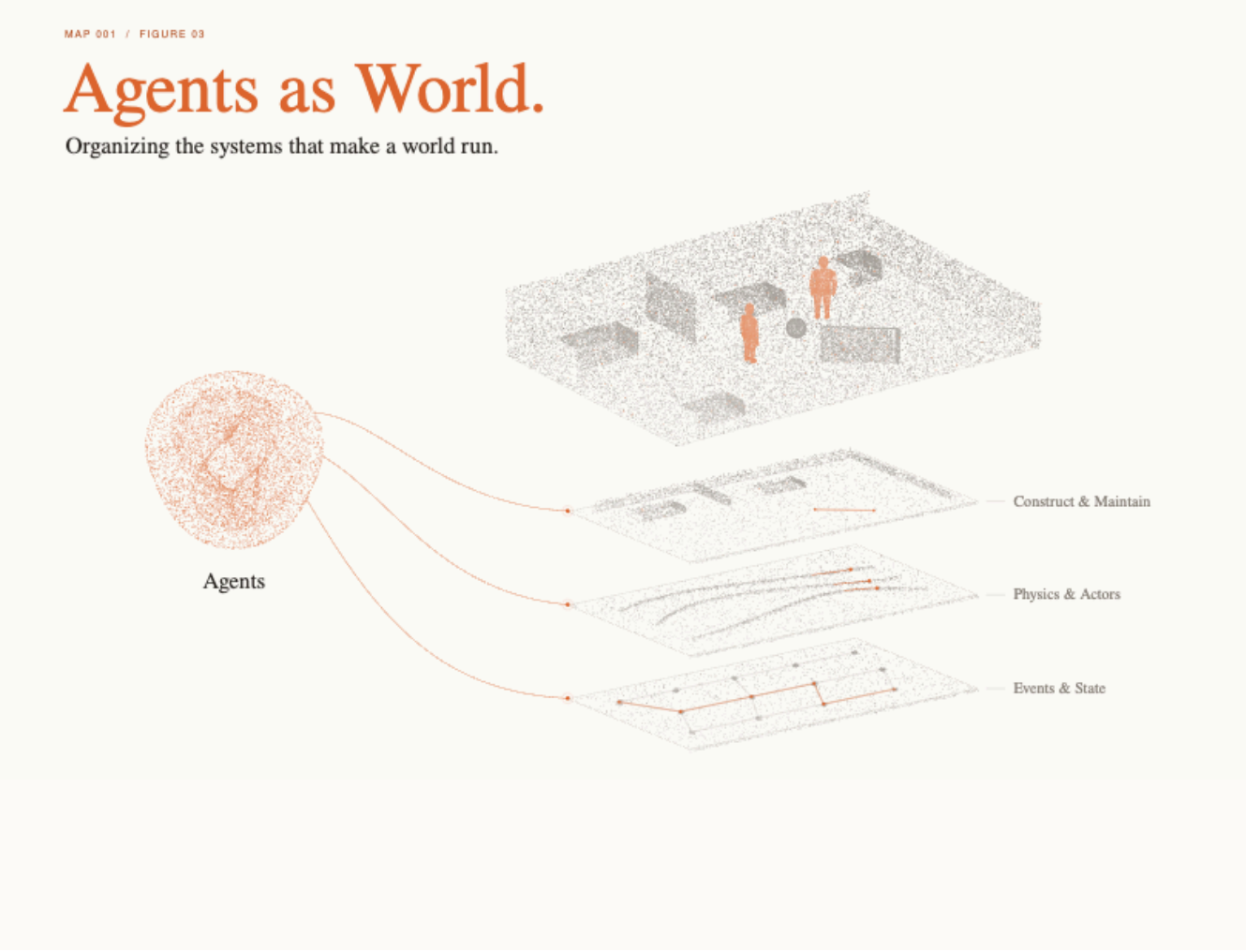
Finally, the world-making process must expose the machinery through which worlds are produced. Changing only finished scenes or trajectories provides little leverage over the conditions of future development. Reusable construction procedures, process models, observation interfaces, event rules, human-dynamics models, and temporal policies must themselves be addressable objects.

The world side of the Second Loop must therefore expose not only states and outcomes, but the processes through which later worlds are **constructed, continued, and revised**.

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3. Agents as World (AAW)

Agents as World (AAW) is our proposed organization for the computational world of the Second Loop. It does not claim that agents replace physics, state, or simulation tools. Its claim is organizational: agents can form the persistent control structure through which a **World Process** is constructed, operated, inspected, and revised, while specialized substrates continue to determine executable consequences.



Why agents? The answer is partly ontological and partly technological.

In human-centered physical intelligence, part of the world is already agentic. People are persistent causal participants, not ~~annotations on a physical scene~~. Their beliefs, memories, intentions, routines, and relationships affect how the world subsequently unfolds. A computational world that treats these dynamics as first-class therefore already contains agents as part of what it represents.

At the same time, computational agents are moving from tool users toward **world operators**. Until recently, the machinery required to construct and operate a computational world was largely assembled from outside it: developers authored scenes, wrote event logic, configured simulators, connected policies, and maintained interfaces across specialized systems. Frontier agents can increasingly operate across these substrates themselves—authoring environments, invoking simulators and renderers, modifying programs, inspecting state, coordinating other agents, and acting through embodied controllers. [15] [16] [17] [18]

The important shift is not any single demonstration. It is the emergence of a common control abstraction across systems that were previously organized separately. Physics engines, renderers, databases, event systems, process models, and robot controllers remain specialized execution substrates. Agents need not reproduce their functions internally; they can organize how those functions are composed and used.

> THE TECHNOLOGICAL PREMISE
>
> Agents do not need to simulate everything. They need to organize the systems that make a world run.

Most agentic simulations place agents **inside** a world. AAW also places agents on the other side of that boundary: as part of the programmable machinery through which the world is made and continued.

This gives agents two distinct roles. **Actor Agents** participate within the world from local perspectives. **World Agents** construct, operate, and revise the processes through which that world unfolds. A shared **World Kernel** preserves authoritative state and executable consequences.

The resulting lifecycle is deliberately process-oriented: **instantiate the world, let it unfold, revise how later worlds are organized, and retain what was learned about making them.**

3.1 Actor Agents: local perspectives that persist

An **Actor Agent** represents a participant whose behavior depends on a local perspective rather ~~than omniscient~~ world state. The abstraction is general: an actor may be a person, robot, worker, visitor,

animal, or another decision-making participant.

What matters is persistence. Identity, goals, beliefs, attention, memories, capabilities, and relationships can carry forward, allowing earlier interactions to change later conditions.

Actor Agents must generate more than dialogue. They may move, wait, ignore an interruption, collaborate, disagree, change a plan, demonstrate a behavior, or create new constraints for other participants. But their actions remain bounded by observation and embodiment: an actor should not know an event it could not observe, nor complete a physical action merely because a language model described it.

The general abstraction keeps AAW applicable across domains. For the class of physical intelligence we care about most, however, one subclass deserves explicit treatment: the human.

3.2 Human Agents: persistent human dynamics

A **Human Agent** is not merely an NPC with a longer prompt. It is a model of a persistent participant whose state can influence—and be influenced by—the physical agent over time. Its specification may contain relatively stable characteristics, while its runtime state carries current goals, beliefs, attention, memories, relationships, routines, preferences, and private observations.

The central modeling target is **interaction dynamics**. The question is not only what a person may believe or intend now, but how interaction changes the state from which that person participates later. An interruption may alter attention; repeated behavior may establish a routine; an unwanted approach may change later willingness to engage.

Human Agents must therefore remain physically situated and epistemically local. What a person can observe depends on position, attention, available signals, and shared history. Proposed physical behavior must pass through the same execution substrate as any other action.

They must also remain non-oracular. A simulated preference, correction, demonstration, or emotional response is internally generated evidence—~~not ground truth~~ about a real person's private state. A Human Agent is a **hypothesis about a participant**, to be calibrated against external observations rather than optimized into compliance.

This is what distinguishes persistent human dynamics from u~~sing people only as data, supervision, or feedback.~~ The person is part of the causal world whose evolution helps determine what interaction becomes possible next.

3.3 World Agents: construct, operate, revise

World Agents are responsible for the non-actor world across three stages: **construction, operation, and revision.** Their purpose is not merely to generate scenes, but to keep the mechanisms through

which the world behaves accessible as the World Process continues.

During **construction**, World Agents interpret a world specification, select or create assets, compose layouts, configure physical properties, instantiate processes, and assemble the tools required for execution. They may operate scene generators, physics engines, Blender, code, procedural pipelines, or validation utilities; no single simulator backend defines AAW.

During **operation**, their responsibility continues. They maintain semantic process state, schedule events, manage dependencies, react to actor actions, and coordinate tools. A world is not complete when its geometry loads. Appliances run, visitors arrive, tasks are interrupted, objects change state, and processes continue outside the learner's current focus.

During **revision**, World Agents use new evidence, discrepancies, or changed requirements to modify scene specifications, process models, physical parameters, event rules, or the programs through which later worlds are constructed and operated. A revision may affect one world instance or become a reusable mechanism shared across many later worlds.

This separation is also what makes the architecture scalable. Expensive agent reasoning need not occur at every physics step or for every object. Agents can author and supervise reusable processes while ordinary programs execute them until a new decision, interruption, or revision is required.

3.4 The World Kernel: revision without rewriting history

A revisable world creates a fundamental tension: the processes through which the world operates must be changeable without making ~~already-observed history negotiable~~.

AAW therefore places Actor and World Agents above a shared **World Kernel**: the authoritative layer for public state, observation boundaries, event ordering, tool execution, snapshots, branching, and replay.

> Agents propose. Tools execute. The world commits.

An agent may decide that a person begins walking, a door should open, or a process should start. The corresponding simulator, program, or declared abstraction determines the resulting state transition. Agents may revise the mechanisms used by later world versions; they may not retroactively rewrite a consequence already committed to the current history.

A compact implementation view is

$$\mathcal{W}_\phi = (G_\phi, B_\phi, C_\phi), \tag{1}$$

where G_ϕ is the graph of Actor and World Agents together with their roles, tools, and communication permissions; B_ϕ is the World Kernel and its executable backends; and C_ϕ coordinates causal time and activation. The physical agent under development participates in the resulting world rather than occupying an omniscient position within the world-building system.

The Kernel is therefore more than an engineering convenience. It is an **epistemic boundary** for the Second Loop. Simultaneous actions require arbitration; agents cannot consume state they could not observe; snapshots preserve pending events, active processes, memories, and backend state; reproducible branches pin relevant mechanisms and randomness.

Future mechanisms may be revised. **Committed evidence may not be rewritten.**

3.5 Persistent time, selective detail

A world that supports long-horizon development must preserve the consequences of its history without paying fine-grained computational cost for every moment. AAW therefore separates the **continuity of world time** from the **resolution of computation** used to represent it.

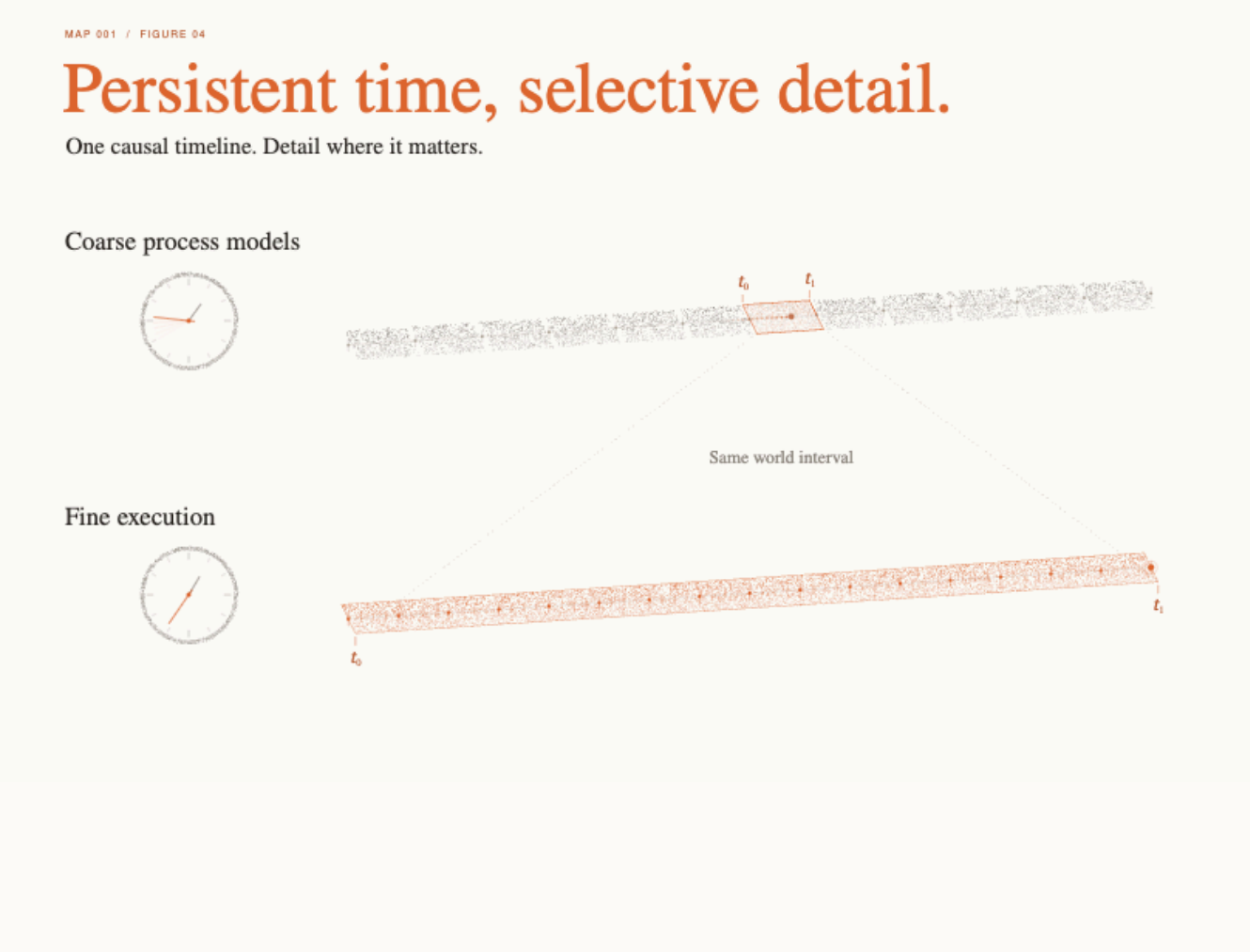
The world has one causal timeline with physical meaning. A three-second action occupies three seconds of world time regardless of how many model calls or physics substeps are needed to compute it. Likewise, a quiet hour need not receive the same dense reasoning as a three-minute human–robot interaction.

Coarse process models can advance routine periods while preserving active conditions, memory, and opportunities for interruption. When interaction becomes dense, the relevant interval expands into fine execution: speech, motion, attention, interruption, cooperation, and physical response unfold at the granularity required by the situation.

Resolution changes dynamically. An unexpected sound, a robot-initiated approach, or a process failure can open a detailed interval even when the coarse plan predicted no interaction. After fine execution, realized outcomes update what follows.

A strict boundary separates planned futures from committed history. Refinement can resolve what has not happened yet; it cannot rewrite a fact another participant has already observed.

The objective is **long-horizon continuity without uniform computational detail**.



3.6 Inheritance: retain how worlds are made

Continual development should retain more than finished scenes and trajectories. It should retain what has been learned about **how useful future worlds are made**.

Inherited artifacts may include observation interfaces, interruption models, asset-construction programs, calibrated actor models, process templates, evaluators, temporal policies, or procedures for assembling and testing world variants. What matters is leverage across time: something learned while constructing or operating one world changes what later worlds can express.

> Memory retains what happened. Inheritance retains how future worlds can be made.

This creates several distinct levels of change. An actor changing its mind is a state transition. Revising the model through which such changes are produced is a World-Process update. Revising the procedure

used to construct, test, or select such models changes the infrastructure through which later developmental situations are produced.

These levels should remain distinct. The learner and the world also have different responsibilities: the learner seeks useful behavior, while the world seeks better coverage and a better account of relevant interactions. Making simulated people more ~~compliant~~ is not progress in modeling human dynamics; increasing the frequency of a difficult scenario is a curriculum intervention, not evidence that the scenario is common in reality. [10]

Development becomes cumulative when what is retained at one level changes the conditions available at the next. Better world-making mechanisms change what the learner can experience; a changed learner changes which interactions matter; new external evidence can revise both.

Inheritance is how the Second Loop carries this leverage forward.

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4. Closing the loops: development across worlds

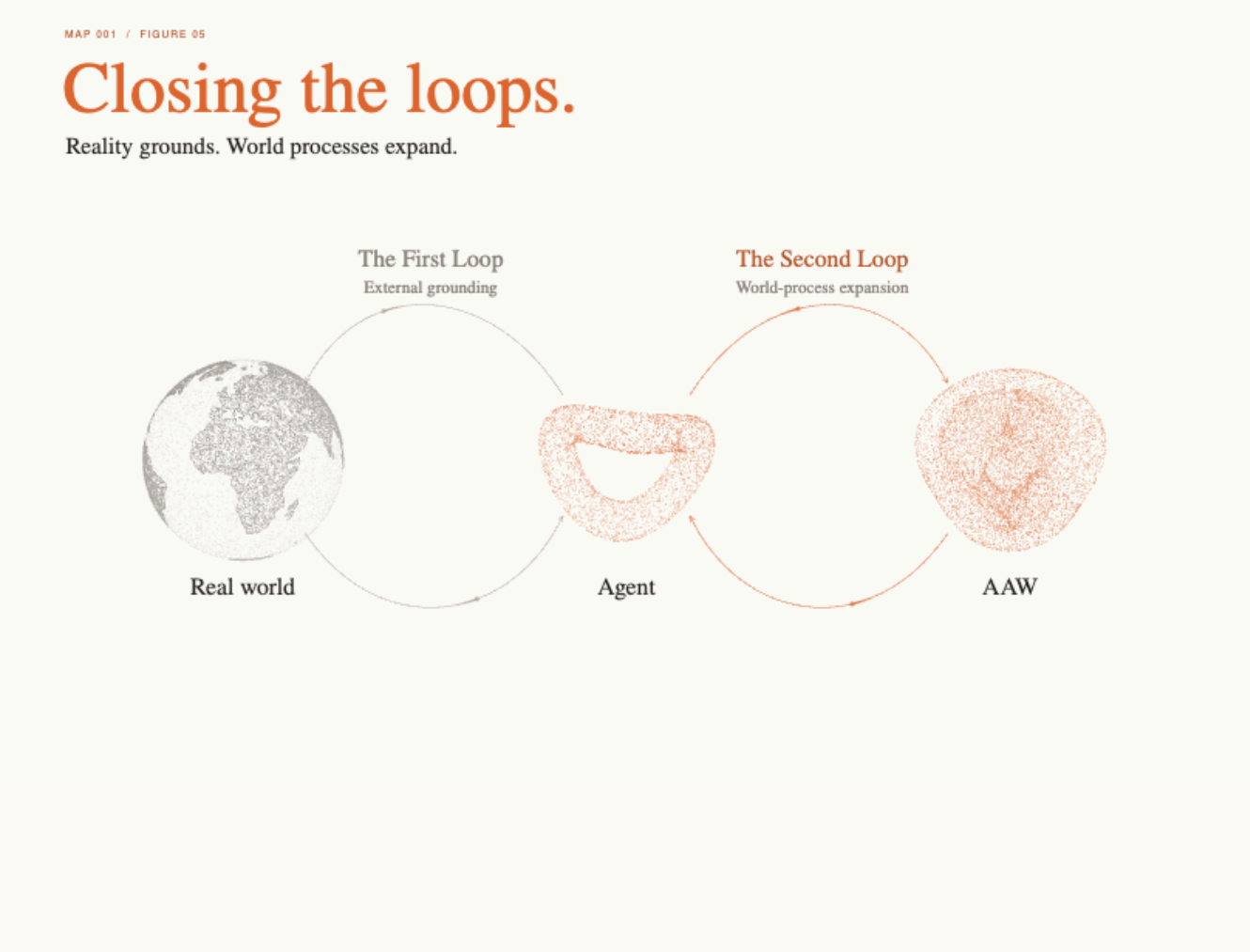
Real2Sim2Real as an epistemic bridge

The First and Second Loops are not successive stages of a pipeline. They are two coupled regimes in which machine development unfolds.

Reality supplies what computation cannot manufacture: embodied consequence, unanticipated interaction, persistent human response, and histories that were not authored for the learner. The Second Loop supplies what reality cannot freely provide: branching, compression, replay, controlled variation, and the ability to revise the machinery through which later experience is constructed.

Their value lies in this asymmetry.

The Second Loop remains meaningful only insofar as its worlds continue to be corrected by what happens outside them. Reality, in turn, becomes more informative when individual experiences can be reconstructed, contrasted, and extended beyond the single trajectory that actually occurred. Real2Sim2Real is therefore not simply a transfer pipeline between simulation and hardware. It is an epistemic coupling between lived experience and computational possibility.



For Machine Aliveness, this coupling is inseparable from deployment. The phenomena we care about most—continuity, adaptation, evolving expectations, persistent relationships—do not fully exist in isolated laboratory episodes. They emerge when machines remain in the world long enough for their past actions to become part of the conditions of their future.

This is where the relationship between Machine Aliveness and Luvbotics becomes structurally important. Machine Aliveness studies the principles by which embodied machines may continue to develop within physical and social worlds; Luvbotics places such machines into the environments where those worlds actually unfold. Deployment exposes dynamics that no designed benchmark can completely anticipate. Research turns those encounters into better abstractions, computational worlds, and developmental mechanisms.

The relationship is therefore not research followed by product. **It is a shared loop between machines that live in the world and systems that learn how to represent, extend, and revise that life.**

Meet the machine entering the world →

Machine intelligence asks what a machine can do. Machine aliveness asks what it can become. **The Second Loop gives that becoming another world in which to unfold—without severing it from the**

one that makes it real.

Research context

The Second Loop builds on a long-standing view of intelligence as something that develops through extended interaction rather than isolated task completion. Developmental robotics emphasizes the role of physical embodiment and interaction with both environments and other agents, while lifelong robot learning asks how experience accumulated across a machine's lifetime can shape what it learns next. [19] [20] Our concern begins from the same temporal premise, but shifts attention toward the **world-side infrastructure** required when development extends across persistent physical and social histories.

Simulation, world models, and open-ended learning make another part of this infrastructure computational. Simulation separates the scale of learning experience from direct physical execution; learned world models make aspects of environmental dynamics available for prediction and imagination; and systems such as POET show that environments themselves can participate in an open-ended learning process. [3] [4] [11] These works move the world from a fixed backdrop toward an object that can be generated, searched, and adapted.

Persistent social agents expose a different requirement. Generative Agents and Virtual Community model participants whose memories, activities, and interactions continue over time, while generalist embodied systems increasingly ground behavior in human demonstrations, instructions, and interactive feedback. [5] [6] [13] [14] For machines that repeatedly encounter the same people, these threads meet: human state is not only supervision for the learner, but part of the continuing dynamics of the world the learner returns to.

At the same time, the machinery surrounding intelligent systems is becoming increasingly programmable by agents themselves. Self-modifying systems such as Darwin Gödel Machine and HyperAgents make code, scaffolds, evaluators, and improvement procedures explicit objects of change, while newer agentic systems increasingly operate across software, 3D world construction, simulation, and embodied control. [1] [2] [8] [9] [15] [16] [17] [18] Related work has also argued that unexpected physical experience can require revising both an embodied actor and its computational account of the world. [12] Together, these developments suggest that agents can increasingly participate not only **inside** computational worlds, but in the machinery through which those worlds are constructed and revised.

The Second Loop is our attempt to name the systems object that emerges at this intersection. A **World Process** carries participants, processes, and consequences forward through time. **Agents as World** makes the organization of that process itself programmable. The **World Kernel** preserves the boundary between revisable world-making machinery and committed causal history.

The resulting perspective is not simply to place a learning agent inside a richer simulator. It is to make the continuing world—and increasingly, the machinery by which that world is made—part of the computational substrate through which a machine develops.

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